

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	XXXXXX
Project Title	AI-Powered Lead Generation System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

AI-Powered Lead Generation System

The **AI-Powered Lead Generation System** is an Artificial Intelligence-based system developed to identify, analyze, and prioritize potential customers using Machine Learning and predictive analytics. The project aims to improve the lead generation process by analyzing customer information, predicting lead quality, assigning lead scores, and providing intelligent recommendations through an interactive web application. This AI-driven approach helps businesses improve sales efficiency, increase conversion rates, and make better data-driven decisions.

Project Overview	
Objective	The primary objective of the project is to develop an AI-powered lead generation system that identifies potential customers, predicts lead quality, and provides lead scoring and recommendations to improve sales performance and customer engagement.

Scope	The project focuses on customer data analysis, machine learning-based lead prediction, automated lead scoring, and integration with a Streamlit web application. The system analyzes lead information, identifies high-potential prospects, and provides real-time insights for sales teams.
Problem statement	
Description	Businesses often struggle to identify high-quality leads due to time-consuming manual processes, inefficient customer analysis, and difficulty in predicting customer conversion. Existing lead generation methods may not provide accurate insights for prioritizing potential customers.
Impact	Addressing these challenges will reduce manual effort, improve lead quality, increase conversion rates, enhance sales decision-making, and provide businesses with an efficient AI-based approach for customer acquisition
Proposed Solution	
Approach	Develop an AI-powered system using Machine Learning and predictive analytics to analyze customer data, identify potential leads, generate lead scores, and provide recommendations through a Streamlit web application.
Key Features	• AI-based lead identification and prediction. • Automated lead scoring based on customer data. • Intelligent recommendations for high-potential prospects. • Interactive Streamlit web interface. • Real-time lead analysis and insights. • Efficient data processing and predictive analytics.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications	Intel Core i5/i7 Processor or Google Colab with NVIDIA T4 GPU
Memory	RAM specifications	Minimum 8 GB RAM
Storage	Space for datasets, trained models, and project files	256 GB SSD or higher
Software		
Frameworks	Python frameworks	Streamlit , Flask (optional)
Libraries	Python Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Visual Studio Code (VS Code), Jupyter Notebook, Google Colab

Data		
Dataset	Source, size, format	Customer/Lead dataset in CSV format containing customer details, interactions, and historical lead information used for training and testing the machine learning model.